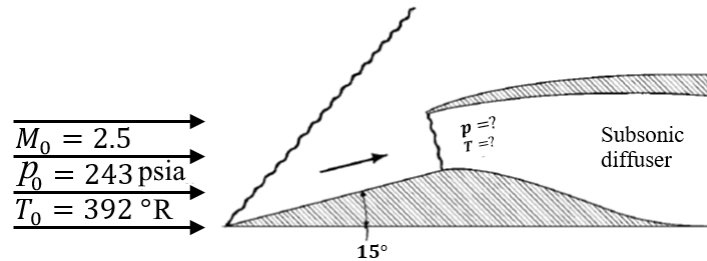
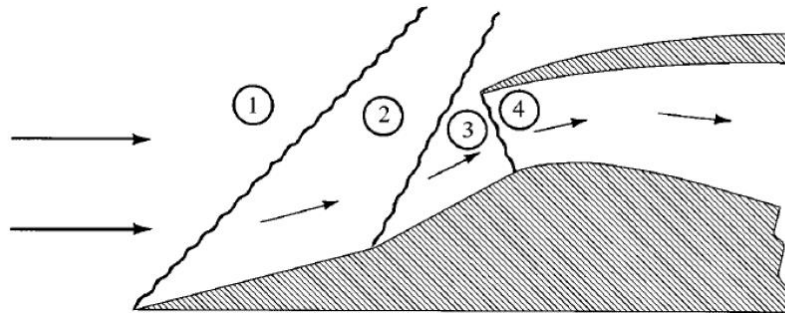


Homework 8

1. (3 points) Consider the flow over a cone of 20° half cone angle, at 0° angle of attack.
 - a. For a free stream Mach number of $M_\infty = 3$ (assume standard sea level conditions), evaluate the shock wave angle, p , T , ρ , and M immediately behind the shock wave and on the cone surface.
 - b. Evaluate the minimum free stream Mach number for the flow to remain attached.
 - c. Evaluate the drag coefficient for the Mach number range of $1.5 < M_\infty < 5.5$. The drag coefficient is defined as $C_D = D/q_\infty A_b$, where A_b is the area of the base of the cone. Assume p_b is equal to the free stream pressure.
2. The two figures below show the intakes of a supersonic airplane. The pressure and temperature of the ambient flows are 243 psia and 392 degrees R. $M_0 = 2.5$
 - a- Find T , P , M and change in entropy just after the normal shock in the first figure.

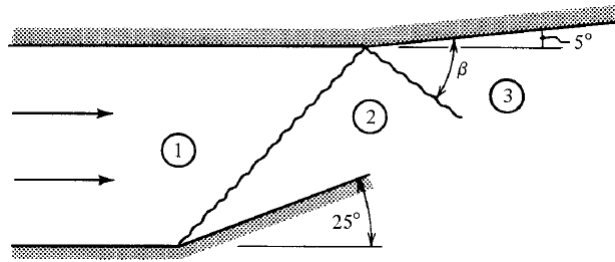


- b- If the wedge with 15 degree is replaced with two wedges with 7 and 8 degrees, as figure below, find T , P , M and change in entropy in region 4.



- c- Report drop in P_0 for both cases a and b and make a conclusion.

3. Flow with $M=3.3$ approaches the below configuration. Find: T_3/T_1 , p_3/p_1 , M_3 and β .



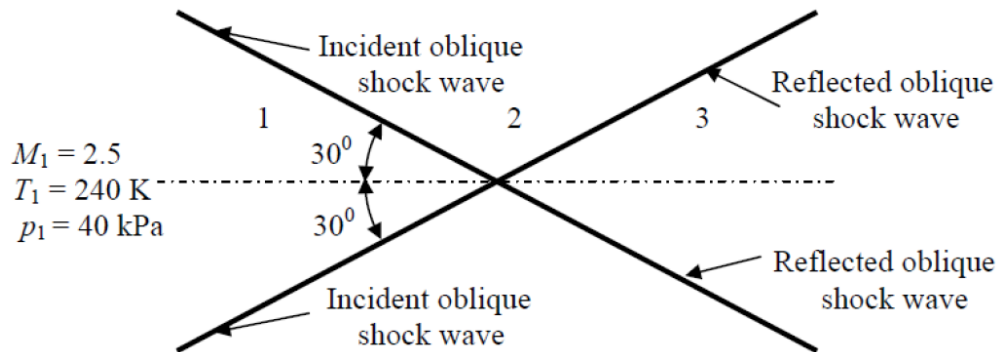
4. Two oblique shocks with $\theta=30$ interact with each other's. Find:

a- P, T and M in region 2.

b- P, T and M in region 3.

c- Angle of the reflected shocks.

Hint: The reflected shocks make the flow again horizontal.



5- For the below geometry, find C_p and C_d .

